Due Date: 11:59 pm on 3/28/2026
Instructor: Dr. Tutku Karacolak

ECE 370 Electromagnetic Fields and Waves

HOMEWORK #5

I expect you to clearly show me how you solved these problems – an orderly and visible presentation of your thoughts and process toward solution. I want to easily see that you understand what you are doing.

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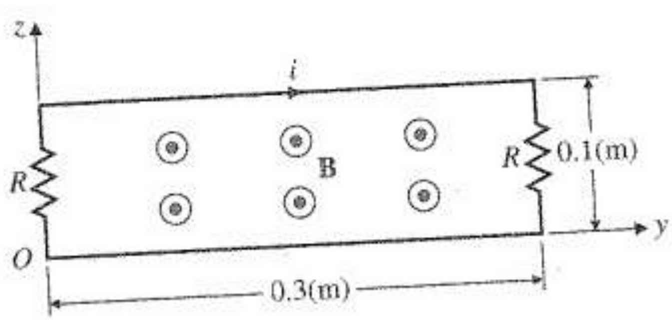
Problem	Course outcome	Grade
1	1-a, 1-c	/30
2	1-a, 1-d	/30
3	1-a, 1-c, 1-d	/40
	TOTAL:	/100

1- A region of space with $\mu = \mu_0$ has a magnetic field $\vec{H} = 4\hat{a}_x + 8\hat{a}_y + 3\hat{a}_z$ (A/m). A 1 Coulomb charged particle is injected into the region with a velocity $\vec{v} = 20\hat{a}_y$ (m/s). If you want the velocity of this particle to remain exactly $\vec{v} = 20\hat{a}_y$ (m/s), what electric field must be added to the region to counteract the magnetic field?

2- The circuit below is situated in a magnetic field

$$\vec{B} = \hat{a}_x 3\cos\left(5\pi 10^7 t - \frac{1}{3}\pi y\right) \quad (\mu T).$$

Assuming $R = 15\ \Omega$, find the current i.



3- Consider the two coplanar conductors (infinitely long wire and the conducting rectangular loop) with the given coordinate system. A current I is flowing on the long wire.

(a) Find the induced voltage in the rectangular loop if the loop is stationary and the current $I = I_0 \cos(\omega t)$. (15 pts)

(b) Repeat part (a) if the current is I_0 and the loop moves away from the long wire with a velocity $\vec{u} = \hat{a}_x u_0$. (15 pts)

(c) Repeat part (a) if the current is $I = I_0 \cos(\omega t)$, and the loop moves away from the long wire with a velocity $\vec{u} = \hat{a}_x u_0$. (10 pts)